Business Requirements Document (BRD)

Shopping Cart

Project: Project AI Mastery

Module ID: BRD-MOD-003

Version: 1.0

Status: Draft

Priority: Critical

Business Domain: Shopping Cart Management

1. Module Overview

The Shopping Cart Module provides users with a centralized view of all products selected for purchase during the current authenticated session.

It enables users to:

- Review selected products
- Verify product information
- Remove products from the cart
- Continue shopping
- Proceed to checkout
- Confirm pricing before purchase

The Shopping Cart acts as an intermediate stage between product selection and checkout, ensuring users can review and modify their selections before entering the purchasing workflow.

2. Business Background

In an e-commerce application, users often select multiple products before making a purchasing decision. The Shopping Cart provides a temporary holding area where selected products can be reviewed, validated, modified, or removed before checkout.

The Cart reduces purchasing errors, improves transparency, and gives users control over their purchase decisions.

3. Business Need

The organization requires a Shopping Cart that:

- Displays all selected products.
- Preserves product selections during the active session.
- Allows removal of products.
- Displays accurate product information.
- Displays correct pricing.
- Supports navigation back to the Inventory page.
- Allows users to continue to the checkout process.
- Maintains synchronization with the Inventory module.

4. Business Objectives

Objective ID	Business Objective
BO-CART-001	Display all selected products.
BO-CART-002	Allow users to review product information before checkout.
BO-CART-003	Allow users to remove products from the cart.
BO-CART-004	Allow users to continue shopping.
BO-CART-005	Allow users to proceed to checkout.
BO-CART-006	Maintain synchronization between Inventory and Cart.
BO-CART-007	Preserve cart contents during the authenticated session.
BO-CART-008	Display accurate pricing for all selected products.
BO-CART-009	Provide a consistent and intuitive user experience.
BO-CART-010	Support secure navigation to Checkout.

5. Scope

In Scope

- Shopping Cart page
- Selected product list
- Product information
- Product quantity (displayed as provided by the application)
- Product description
- Product price
- Remove button
- Continue Shopping button
- Checkout button
- Header
- Shopping Cart badge synchronization

Out of Scope

- Quantity modification
- Coupon codes
- Promotional discounts
- Shipping estimation
- Tax calculation
- Save for Later
- Wishlist
- Gift wrapping
- Inventory reservation
- Multi-cart support

Note: *SauceDemo displays a quantity column, but it is fixed at* ***1*** *for each selected product. Quantity editing is not supported and is therefore out of scope.*

6. Stakeholders

Stakeholder	Responsibility
Product Owner	Defines cart business requirements
Business Analyst	Documents business requirements
UI/UX Designer	Designs cart interface
Development Team	Implements cart functionality
QA Team	Validates cart behavior
End Users	Review products before checkout

7. Assumptions

The following assumptions apply:

- User has successfully authenticated.
- User has access to the Inventory page.
- Product information exists.
- Shopping Cart service is operational.
- Session is active.
- Product pricing is available.
- Checkout module is available.

8. Dependencies

The Shopping Cart Module depends upon:

- Login Module
- Inventory Module
- Product Repository
- Session Management
- Checkout Module

9. User Roles

User Role	Access
Standard User	Full cart functionality
Locked User	No access
Problem User	Cart accessible with simulated issues for testing
Performance User	Cart accessible with simulated delays
Error User	Cart accessible with simulated application errors
Visual User	Cart accessible for visual validation

10. Module Overview

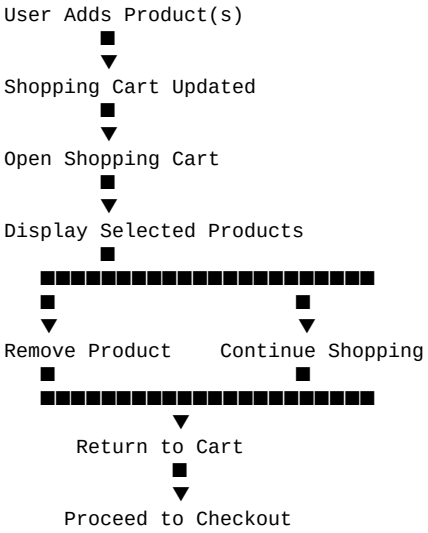
The Shopping Cart serves as the user's review screen before checkout.

Users can:

- View selected products.
- Verify product information.
- Remove unwanted products.
- Continue shopping.
- Proceed to checkout.

The Shopping Cart shall always display the current state of the user's selected products.

11. High-Level Business Flow



12. Success Criteria

The Shopping Cart Module shall be considered successful when:

- Only selected products are displayed.
- Product information matches the Inventory page.
- Product prices are accurate.
- Users can remove products successfully.
- Continue Shopping returns users to the Inventory page.
- Checkout navigation works correctly.
- Shopping Cart badge remains synchronized.
- Cart contents persist throughout the active session.
- Navigation occurs without data loss or inconsistencies.

Next: Part 2 – Functional Requirements

Excellent. The Cart module is smaller than Inventory but is still a **critical business module** because it validates the user's selected products before checkout. The Functional Requirements below are written in an enterprise BRD style and map directly to future test cases and automation.

13. Functional Requirements

FR-CART-001 – Display Shopping Cart Page

Requirement ID

FR-CART-001

Requirement Name

Display Shopping Cart Page

Priority

Critical

Description

The system shall display the Shopping Cart page when the user selects the Shopping Cart icon from the Inventory page.

Preconditions

- User is authenticated.
- Active session exists.

Trigger

User selects the Shopping Cart icon.

Functional Behaviour

The system shall:

- Open the Shopping Cart page.
- Display the page header.
- Display the selected products.
- Display the Continue Shopping button.
- Display the Checkout button.

Postconditions

The Shopping Cart page is fully loaded and ready for interaction.

Acceptance Criteria

- Cart page opens successfully.
- No application errors occur.
- Cart contents are displayed correctly.

FR-CART-002 – Display Selected Products

Requirement ID

FR-CART-002

Description

The system shall display only those products that the user has added to the shopping cart.

Functional Behaviour

- Display selected products only.
- Do not display unselected products.
- Preserve the order of products as maintained by the application.

Acceptance Criteria

Only selected products are displayed.

FR-CART-003 – Display Product Information

Description

Each selected product shall display complete product information.

Product Information

- Product Name
- Product Description
- Product Price
- Product Quantity
- Remove Button

FR-CART-004 – Display Product Quantity

Description

Each product shall display its quantity.

Functional Behaviour

For the current SauceDemo implementation:

- Quantity is displayed.
- Quantity value is fixed at **1**.
- Quantity cannot be modified.

Acceptance Criteria

Quantity is displayed correctly for every product.

FR-CART-005 – Display Product Price

Description

Each product shall display its current selling price.

Acceptance Criteria

Displayed price matches the Inventory page.

FR-CART-006 – Product Details Navigation

Description

Users shall be able to navigate to the Product Details page by selecting the product name.

Acceptance Criteria

Correct Product Details page is displayed.

FR-CART-007 – Remove Product

Priority

Critical

Description

Users shall be able to remove products from the Shopping Cart.

Functional Behaviour

Upon selecting **Remove**:

- Product is removed.
- Shopping Cart badge updates.
- Inventory page reflects updated state.

Acceptance Criteria

Selected product is removed successfully.

FR-CART-008 – Continue Shopping

Description

Selecting **Continue Shopping** shall navigate the user back to the Inventory page.

Acceptance Criteria

Inventory page opens while preserving remaining cart contents.

FR-CART-009 – Proceed to Checkout

Priority

Critical

Description

Selecting **Checkout** shall navigate the user to the Checkout Information page.

Preconditions

At least one product exists in the cart.

Acceptance Criteria

Checkout Information page opens successfully.

FR-CART-010 – Shopping Cart Badge Synchronization

Description

The Shopping Cart badge shall remain synchronized with the products displayed in the Shopping Cart.

Acceptance Criteria

Badge count equals the total number of selected products.

FR-CART-011 – Cart Persistence

Description

Selected products shall remain in the Shopping Cart throughout the authenticated session.

Acceptance Criteria

Refreshing the page does not remove selected products.

FR-CART-012 – Product Data Consistency

Description

Product information displayed in the Shopping Cart shall remain consistent with the Inventory and Product Details pages.

Acceptance Criteria

No inconsistencies exist between modules.

FR-CART-013 – Empty Shopping Cart

Description

If all products are removed, the Shopping Cart shall display an empty cart state.

Functional Behaviour

- No products displayed.
- Continue Shopping remains available.
- Checkout behavior follows business rules.

Note: *SauceDemo does not display a dedicated empty-state message. The page simply shows no products while retaining the action buttons.*

FR-CART-014 – Header Display

Description

The Shopping Cart page shall display the standard application header.

Components

- Hamburger Menu
- Application Logo
- Shopping Cart Icon

FR-CART-015 – Shopping Cart Title

Description

The page shall display the title **"Your Cart"**.

FR-CART-016 – Continue Shopping Button

Description

The Continue Shopping button shall always be available on the Shopping Cart page.

FR-CART-017 – Checkout Button

Description

The Checkout button shall be available to allow users to proceed to the checkout process.

FR-CART-018 – Responsive Layout

Description

The Shopping Cart page shall adapt to supported screen sizes while maintaining readability and usability.

Supported devices:

- Desktop
- Laptop
- Tablet
- Mobile

FR-CART-019 – Browser Navigation

Description

Using browser Back or Forward navigation shall not result in inconsistent cart contents or application errors.

FR-CART-020 – Session Validation

Description

The Shopping Cart shall only be accessible while the user's authenticated session remains valid.

If the session becomes invalid or expires, the user shall be redirected to the Login page before further interaction.

Functional Requirement Summary

Requirement ID	Requirement
FR-CART-001	Display Shopping Cart Page
FR-CART-002	Display Selected Products
FR-CART-003	Display Product Information
FR-CART-004	Display Product Quantity
FR-CART-005	Display Product Price
FR-CART-006	Product Details Navigation
FR-CART-007	Remove Product
FR-CART-008	Continue Shopping
FR-CART-009	Proceed to Checkout
FR-CART-010	Shopping Cart Badge Synchronization
FR-CART-011	Cart Persistence
FR-CART-012	Product Data Consistency
FR-CART-013	Empty Shopping Cart
FR-CART-014	Header Display
FR-CART-015	Shopping Cart Title
FR-CART-016	Continue Shopping Button
FR-CART-017	Checkout Button
FR-CART-018	Responsive Layout
FR-CART-019	Browser Navigation

Requirement ID	Requirement
FR-CART-020	Session Validation

14. Business Rules

BR-CART-001 – Authenticated Access Only

Rule ID: BR-CART-001

Only authenticated users with a valid session shall be permitted to access the Shopping Cart.

Priority: Critical

BR-CART-002 – Display Selected Products Only

The Shopping Cart shall display only those products that have been explicitly added by the authenticated user.

Products not selected by the user shall not appear in the Shopping Cart.

BR-CART-003 – Cart Synchronization

The Shopping Cart shall always remain synchronized with the Inventory module.

Any product added or removed from the Inventory page shall be immediately reflected in the Shopping Cart.

BR-CART-004 – Accurate Product Information

Each product displayed in the Shopping Cart shall present accurate and consistent information, including:

- Product Name
- Product Description
- Product Price
- Quantity
- Product Identifier

The displayed information shall match the Inventory and Product Details pages.

BR-CART-005 – Quantity Display

Each selected product shall display its quantity.

For the current SauceDemo implementation:

- Quantity is fixed at **1**
- Quantity is read-only
- Users cannot modify quantity

BR-CART-006 – Product Pricing

The displayed product price shall match the current selling price maintained by the application.

Prices shall:

- Include the currency symbol.
- Display proper decimal formatting.
- Be consistent across all application modules.

BR-CART-007 – Product Details Navigation

Selecting the product name shall navigate the user to the corresponding Product Details page.

BR-CART-008 – Remove Product

Selecting **Remove** shall remove only the selected product from the Shopping Cart.

No other product shall be affected.

BR-CART-009 – Shopping Cart Badge

The Shopping Cart badge shall display the total number of products currently present in the cart.

The badge shall:

- Increase when products are added.
- Decrease when products are removed.
- Disappear when the cart becomes empty.

BR-CART-010 – Continue Shopping

Selecting **Continue Shopping** shall return the user to the Inventory page while preserving the current Shopping Cart contents.

BR-CART-011 – Proceed to Checkout

Selecting **Checkout** shall navigate the user to the Checkout Information page.

This action shall preserve all selected products for the checkout process.

BR-CART-012 – Empty Cart Behavior

When all products have been removed:

- The Shopping Cart shall display no product entries.
- Continue Shopping shall remain available.
- Checkout button behavior shall follow the application's implemented logic.

Note: *SauceDemo currently allows navigation to Checkout even when the cart is empty. This is documented as the application's existing behavior rather than a recommendation for production systems.*

BR-CART-013 – Cart Persistence

Shopping Cart contents shall remain available throughout the authenticated session unless:

- Products are removed.
- Session expires.
- User logs out.
- Application state is reset.

BR-CART-014 – Data Consistency

Product information shall remain consistent across:

- Inventory
- Product Details
- Shopping Cart

- Checkout

BR-CART-015 – Independent Product Actions

Removing one product shall not affect:

- Other selected products.
- Product information.
- Remaining cart contents.

BR-CART-016 – Page Refresh Behavior

Refreshing the Shopping Cart page during an active session shall preserve current cart contents.

BR-CART-017 – Browser Navigation

Using browser Back or Forward navigation shall not result in:

- Duplicate products.
- Incorrect badge counts.
- Data inconsistencies.
- Application errors.

BR-CART-018 – Header Consistency

The Shopping Cart shall display the standard application header containing:

- Hamburger Menu
- Application Logo
- Shopping Cart Icon

The header shall remain consistent with other authenticated pages.

BR-CART-019 – Secure Session Enforcement

If the user's session becomes invalid or expires while viewing the Shopping Cart, the application shall prevent further cart interactions and redirect the user to the Login page.

BR-CART-020 – Navigation Integrity

Navigation between Inventory, Shopping Cart, and Checkout shall not alter the cart contents unless explicitly requested by the user.

BR-CART-021 – Product Order Consistency

Products displayed in the Shopping Cart shall maintain the application's defined ordering and shall not be reordered unexpectedly during navigation or page refresh.

BR-CART-022 – Responsive Display

The Shopping Cart page shall maintain usability, readability, and functional consistency across all supported screen sizes and browsers.

BR-CART-023 – Accessibility

All interactive controls on the Shopping Cart page shall support keyboard navigation and assistive technologies in accordance with accessibility standards.

BR-CART-024 – Error Handling

If product information cannot be retrieved or displayed, the application shall present an appropriate user-facing error without exposing internal implementation details.

BR-CART-025 – Cart Integrity

The Shopping Cart shall maintain accurate product information, synchronization, navigation, and checkout readiness throughout the authenticated session.

Business Rule Summary

Rule ID	Business Rule
BR-CART-001	Authenticated Access Only
BR-CART-002	Display Selected Products Only
BR-CART-003	Cart Synchronization
BR-CART-004	Accurate Product Information
BR-CART-005	Quantity Display
BR-CART-006	Product Pricing
BR-CART-007	Product Details Navigation
BR-CART-008	Remove Product
BR-CART-009	Shopping Cart Badge
BR-CART-010	Continue Shopping
BR-CART-011	Proceed to Checkout
BR-CART-012	Empty Cart Behavior
BR-CART-013	Cart Persistence
BR-CART-014	Data Consistency
BR-CART-015	Independent Product Actions
BR-CART-016	Page Refresh Behavior
BR-CART-017	Browser Navigation
BR-CART-018	Header Consistency
BR-CART-019	Secure Session Enforcement
BR-CART-020	Navigation Integrity
BR-CART-021	Product Order Consistency
BR-CART-022	Responsive Display
BR-CART-023	Accessibility
BR-CART-024	Error Handling
BR-CART-025	Cart Integrity

15. User Interface (UI) Requirements

15.1 Shopping Cart Page Layout

UI-CART-001 – Shopping Cart Page Structure

Requirement ID

UI-CART-001

Description

The Shopping Cart page shall present selected products in a structured and user-friendly layout that enables users to review their selections before checkout.

Page Components

1. Header
2. Hamburger Menu
3. Application Logo
4. Shopping Cart Title
5. Cart Information Header
6. Product List
7. Continue Shopping Button
8. Checkout Button
9. Footer

Acceptance Criteria

• All components are displayed correctly.

• Layout is visually consistent.

• Page remains responsive across supported devices.

15.2 Header

UI-CART-002 – Header Section

Description

The standard application header shall be displayed at the top of the Shopping Cart page.

Header Components

Component	Mandatory
Hamburger Menu	Yes
Application Logo	Yes
Shopping Cart Icon	Yes

Acceptance Criteria

• Header is always visible.

• Components are properly aligned.

• Header matches the authenticated application pages.

15.3 Hamburger Menu

UI-CART-003 – Navigation Menu

Description

Selecting the Hamburger Menu shall display the application navigation panel.

Menu Options (SauceDemo)

- All Items
- About
- Logout
- Reset App State

Behaviour

- Opens when selected.
- Closes using the close icon or by selecting a menu item.
- Menu items remain selectable.

Acceptance Criteria

- Menu opens successfully.
- Menu closes successfully.
- Navigation options function correctly.

15.4 Shopping Cart Title

UI-CART-004 – Page Title

Description

The Shopping Cart page shall display the title:

"Your Cart"

Requirements

- Clearly visible.
- Properly aligned.
- Consistent styling.

15.5 Cart Information Header

UI-CART-005 – Cart Information Labels

Description

The Shopping Cart shall display column headers describing the product information.

Columns

Column	Description
QTY	Product Quantity
DESCRIPTION	Product Information

Behaviour

- Column headers remain visible.
- Product information aligns correctly beneath each header.

15.6 Product List

UI-CART-006 – Product Listing

Description

Each selected product shall appear as an individual cart item.

Cart Item Components

- Quantity
- Product Name
- Product Description
- Product Price
- Remove Button

Acceptance Criteria

Only selected products are displayed.

15.7 Product Quantity

UI-CART-007 – Quantity Display

Description

Each product shall display its quantity.

Behaviour

For the current SauceDemo implementation:

- Quantity value is **1**
- Quantity is read-only
- Quantity cannot be edited

Acceptance Criteria

Quantity displays accurately.

15.8 Product Name

UI-CART-008 – Product Name

Behaviour

The product name shall:

- Be visible.
- Be readable.
- Navigate to Product Details when selected.
- Match the selected product.

15.9 Product Description

UI-CART-009 – Product Description

Behaviour

Each product shall display its description.

The description shall:

- Match the product.
- Be readable.
- Be consistently formatted.

15.10 Product Price

UI-CART-010 – Product Price

Behaviour

The displayed price shall:

- Include the currency symbol.
- Display proper decimal precision.
- Match the Inventory page.

15.11 Remove Button

UI-CART-011 – Remove Product Button

Initial State

Visible for every selected product.

Behaviour

Selecting **Remove** shall:

- Remove the selected product.
- Update the Shopping Cart badge.
- Update the displayed product list.
- Preserve remaining products.

Acceptance Criteria

Only the selected product is removed.

15.12 Continue Shopping Button

UI-CART-012 – Continue Shopping

Description

The Continue Shopping button shall return users to the Inventory page.

Behaviour

- Always visible.
- Clickable.

- Preserves current cart contents.

Acceptance Criteria

Inventory page opens without losing selected products.

15.13 Checkout Button

UI-CART-013 – Checkout

Description

The Checkout button shall navigate users to the Checkout Information page.

Behaviour

- Always visible.
- Clickable.
- Preserves selected products.

Acceptance Criteria

Checkout Information page opens successfully.

15.14 Shopping Cart Icon

UI-CART-014 – Shopping Cart Icon

Description

The Shopping Cart icon shall remain visible within the application header.

Behaviour

- Displays current cart badge.
- Badge updates immediately when products are removed.
- Icon remains accessible.

15.15 Footer

UI-CART-015 – Footer

Description

The Shopping Cart page shall display the standard application footer.

Footer Components (SauceDemo)

- Twitter/X
- Facebook
- LinkedIn
- Copyright

Behaviour

- Links are clickable.
- Open the associated external pages.
- Footer remains consistent with other application pages.

15.16 Responsive Layout

UI-CART-016 – Responsive Behaviour

The Shopping Cart page shall support:

- Desktop
- Laptop
- Tablet
- Mobile Browser

Requirements

- No overlapping components.
- Product information remains readable.
- Buttons remain accessible.
- Layout adapts appropriately.

15.17 Empty Cart Display

UI-CART-017 – Empty Cart State

Description

If no products exist in the Shopping Cart, the page shall display an empty product list while retaining the standard page layout.

Current SauceDemo Behaviour

- Header remains visible.
- Cart information headers remain visible.
- Continue Shopping button remains available.
- Checkout button remains available.
- No dedicated empty-cart message is displayed.

Note: *Many production e-commerce applications display an "Your cart is empty" message with a call-to-action. SauceDemo does not implement this behavior, so the BRD documents the application's current state.*

UI Requirements Summary

Requirement ID	Requirement
UI-CART-001	Shopping Cart Page Structure
UI-CART-002	Header Section
UI-CART-003	Hamburger Menu
UI-CART-004	Shopping Cart Title
UI-CART-005	Cart Information Header
UI-CART-006	Product Listing
UI-CART-007	Quantity Display
UI-CART-008	Product Name
UI-CART-009	Product Description
UI-CART-010	Product Price

Requirement ID	Requirement
UI-CART-011	Remove Button
UI-CART-012	Continue Shopping Button
UI-CART-013	Checkout Button
UI-CART-014	Shopping Cart Icon
UI-CART-015	Footer
UI-CART-016	Responsive Layout
UI-CART-017	Empty Cart Display

16. Validation Rules

VR-CART-001 – Selected Products Validation

The Shopping Cart shall display only products that have been added by the authenticated user.

Expected Result

- No unselected products are displayed.
- No duplicate product entries exist.

VR-CART-002 – Product Information Validation

Each cart item shall display:

- Product Name
- Product Description
- Product Price
- Product Quantity
- Remove Button

Expected Result

All required product information is present and accurate.

VR-CART-003 – Quantity Validation

For the current SauceDemo implementation:

- Quantity shall always display **1**.
- Quantity shall be read-only.
- Users shall not be able to edit quantity.

VR-CART-004 – Price Validation

Each product shall display:

- Currency symbol
- Correct decimal precision
- Same price as the Inventory and Product Details pages

VR-CART-005 – Remove Button Validation

The **Remove** button shall:

- Be visible for each product.
- Be enabled.
- Remove only the selected product.

VR-CART-006 – Continue Shopping Validation

The **Continue Shopping** button shall:

- Be visible.
- Be enabled.
- Navigate to the Inventory page.

VR-CART-007 – Checkout Button Validation

The **Checkout** button shall:

- Be visible.
- Be enabled.
- Navigate to the Checkout Information page.

VR-CART-008 – Shopping Cart Badge Validation

The shopping cart badge shall:

- Match the number of products displayed.
- Update immediately after add/remove operations.
- Be hidden when the cart becomes empty.

VR-CART-009 – Header Validation

The Shopping Cart page header shall consistently display:

- Hamburger Menu
- Application Logo
- Shopping Cart Icon

VR-CART-010 – Responsive Layout Validation

The Shopping Cart page shall maintain usability across supported devices without overlapping elements or inaccessible controls.

17. Error Handling

ERR-CART-001 – Product Information Unavailable

If product information cannot be retrieved, the application shall present a user-friendly message without exposing internal implementation details.

ERR-CART-002 – Session Expired

If the user's authenticated session expires while viewing the Shopping Cart:

- Further interaction shall be prevented.
- The user shall be redirected to the Login page.

ERR-CART-003 – Navigation Failure

If navigation to the Inventory or Checkout page fails, the application shall:

- Prevent data corruption.
- Preserve current cart contents where possible.
- Present an appropriate error notification.

ERR-CART-004 – Invalid Product State

If a product no longer exists or becomes unavailable, the application shall prevent inconsistent cart behavior.

Note: *SauceDemo does not implement this scenario; it is included as a production-oriented requirement.*

18. Security Requirements

SEC-CART-001 – Authenticated Access

Only authenticated users with a valid session shall access the Shopping Cart.

SEC-CART-002 – Session Validation

Every Shopping Cart request shall validate the user's active session before processing.

SEC-CART-003 – Unauthorized URL Access

Direct access to the Shopping Cart URL without authentication shall redirect the user to the Login page.

SEC-CART-004 – Session Invalidation

After logout, the Shopping Cart shall no longer be accessible until the user authenticates again.

SEC-CART-005 – Sensitive Information Exposure

The Shopping Cart shall not expose:

- Session identifiers
- Authentication tokens
- Internal server errors
- Database errors
- Stack traces

SEC-CART-006 – Secure Navigation

Navigation between Inventory, Cart, and Checkout shall preserve session security and prevent unauthorized access.

19. Non-Functional Requirements

Performance

NFR-CART-001

Shopping Cart page shall load within **3 seconds** under normal operating conditions.

NFR-CART-002

Add, remove, and navigation actions shall complete within **2 seconds** for 95% of requests under expected load.

Availability

NFR-CART-003

The Shopping Cart service shall maintain **99.9%** availability during scheduled operating hours.

Reliability

NFR-CART-004

The Shopping Cart shall consistently preserve selected products throughout the authenticated session.

Scalability

NFR-CART-005

The Shopping Cart shall support concurrent user sessions without functional degradation.

Maintainability

NFR-CART-006

The Shopping Cart shall support future enhancements such as:

- Quantity editing
- Coupon codes
- Promotions
- Wishlist
- Save for Later

Usability

NFR-CART-007

The Shopping Cart interface shall be intuitive and require minimal user training.

20. Accessibility Requirements

The Shopping Cart shall comply with WCAG 2.1 AA guidelines where applicable.

ACC-CART-001

All interactive elements shall be keyboard accessible.

ACC-CART-002

Logical tab order shall be maintained across all controls.

ACC-CART-003

Visible keyboard focus shall be provided for interactive elements.

ACC-CART-004

Product information and action buttons shall be readable by screen readers.

ACC-CART-005

Text and controls shall meet minimum color contrast requirements.

21. Browser & Device Compatibility

Supported Browsers

Browser	Supported
Chrome (latest 2 versions)	Yes
Microsoft Edge	Yes
Firefox	Yes
Safari	Yes (where supported)

Supported Devices

- Desktop
- Laptop
- Tablet
- Mobile Browser

22. Audit & Logging Requirements

The system shall log the following events:

Event	Log Required
Cart Opened	Yes
Product Added	Yes
Product Removed	Yes
Continue Shopping	Yes
Checkout Initiated	Yes
Session Timeout	Yes

Each log entry should include:

- Event ID
- Timestamp
- User Identifier
- Session Identifier
- Event Type
- Browser Information

23. Data Dictionary

Data Element	Type	Description
Product ID	String	Unique product identifier
Product Name	String	Product title
Product Description	String	Product summary
Product Price	Decimal	Product selling price
Product Quantity	Integer	Number of selected units (fixed at 1 in SauceDemo)
Cart Count	Integer	Total products in cart

24. CRUD Matrix

Entity	Create	Read	Update	Delete
Cart Item	✓	✓	✗*	✓
Product Information	✗	✓	✗	✗
Cart Badge	✓	✓	✓	✓

*Quantity updates are not supported in the current SauceDemo implementation.

25. Risks

Risk ID	Risk	Impact	Mitigation
RSK-CART-001	Cart data not synchronized	Incorrect purchases	Validate synchronization between Inventory and Cart
RSK-CART-002	Product price mismatch	Incorrect order total	Validate pricing consistency
RSK-CART-003	Session expiration during checkout	Interrupted purchase	Implement session handling and re-authentication
RSK-CART-004	Navigation failures	Poor user experience	Comprehensive navigation testing
RSK-CART-005	Browser rendering issues	UI inconsistencies	Cross-browser compatibility testing

26. Acceptance Criteria

The Shopping Cart Module shall be accepted when:

Functional

- Selected products display correctly.
- Product information is accurate.
- Remove functionality works correctly.
- Continue Shopping returns to Inventory.
- Checkout navigation functions correctly.

UI

- Header, footer, and controls render correctly.
- Product layout is consistent.
- Responsive behavior is maintained.

Security

- Authentication is required.
- Session validation is enforced.
- Unauthorized access is prevented.

Performance

- Page load and interactions meet defined performance targets.

Accessibility

- Keyboard navigation works correctly.
- Screen readers can interpret page content.
- Focus indicators are visible.

27. Requirements Traceability Matrix

Business Objective	Functional Requirement	Business Rule	Acceptance Criteria
BO-CART-001	FR-CART-002	BR-CART-002	AC-01
BO-CART-003	FR-CART-007	BR-CART-008	AC-02
BO-CART-004	FR-CART-008	BR-CART-010	AC-03
BO-CART-005	FR-CART-009	BR-CART-011	AC-04
BO-CART-007	FR-CART-011	BR-CART-013	AC-05